

A Qualitative Study of Human Theorizing about Robot Bodily Behavior

Martina Ruocco
University of Manchester
Manchester, UK
martina.ruocco@glasgow.ac.uk

Caroline Jay
University of Manchester
Manchester, UK
caroline.jay@manchester.ac.uk

Bijan Parsia
University of Manchester
Manchester, UK
bijan.parsia@manchester.ac.uk

ABSTRACT

This research investigates the cognitive frameworks individuals use to understand and justify robot behaviour. Participants watched a robot learning how to manipulate objects through unguided experiential learning. Across multiple rounds of try-and-fail, participants were asked to predict the robot’s performance with unfamiliar objects and speculate on its cognitive operations, including reasoning, sensing, and actuation. To assess the potential impact of performance on theorization, participants were divided into two groups: one exposed to a predominantly successful robot, and the other to a predominantly failing robot. The findings reveal intriguing patterns: participants, regardless of their condition, consistently engaged in theorizing about the robot in similar quantities and ways. Those who observed the robot failing more frequently often subsequently predicted success, perhaps due to a desire to offer the robot “another chance”, or because probabilistically they felt success was more likely following several failures.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **HCI theory, concepts and models**; **HCI design and evaluation methods**; *Interaction design*.

KEYWORDS

human-robot interaction, trust, theory of mind

ACM Reference Format:

Martina Ruocco, Caroline Jay, and Bijan Parsia. 2024. A Qualitative Study of Human Theorizing about Robot Bodily Behavior. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3613905.3651078>

1 INTRODUCTION

Over the past decade, extensive work has explored the extent to which we can develop humanely intelligent technologies, leading to the expansion of fields such as cognitive robotics [8, 16, 21, 24]. Developing robots capable of adapting dynamically to the real world, and potentially even learning social cues that provide insight into human intentions and desires [25, 27], would boost their utility in public and private settings [4, 21]. From the standpoint of

application-focused research, cognitive robotics has been explored in various fields: from the more generally applicable human-robot interaction [9, 11, 18], to robotics for healthcare [10, 17], to collaborative industrial robotics [5].

However, improved cognitive capabilities alone are not sufficient to ensure optimal collaboration with humans. Studies suggest that trust plays a fundamental role in achieving this [12–14, 19], and a significant branch of the HRI research area focuses on investigating the nature of trust in human-robot interaction [26]. The question addressed is whether it is possible to make trust a variable that can be monitored and used to fine-tune how robots communicate and act around humans [7, 15, 20]. A well-established qualitative method that can be used to capture the motivations, nature, and extent behind trust placement is Thematic Analysis [3], which has been extensively used in the HCI field in order to extract people’s patterns of reasoning (“themes”) when using technologies [1, 2].

In this paper, we use Thematic Analysis with the aim of understanding the mental framework that humans develop when observing a robotic agent demonstrate its cognitive abilities. The data gathered encompasses various aspects of trust in HRI, from the mundane, such as the level of confidence people have in the robot’s ability to execute manual tasks, to the more fundamental, such as whether people theorize about robotic agents as they would about human agents, and the degree of sophistication to which such theorisation occurs.

Here we test our approach with a relatively simple physical task. Our basic experimental design follows the study of Chen et al. [6] where the participants were asked to play a collaborative game with a robot in which they had to successfully clear the table of a sequence of objects. Human trust was modelled as a latent variable, and it was measured proportionally to the human intervention rate in the task. If the human intervenes – i.e. stops the robot from continuing – that is taken to indicate distrust. Given that having multiple rounds supports monitoring the evolution of a participant’s reasoning, and the predictive task was well-defined, we decided to try to replicate the structure of the game where possible, but to rewrite its specifics to suit a more qualitative investigation.

2 MATERIALS AND METHODS

2.1 Participants

We recruited 12 participants (6 males and 5 females, age: $\mu = 28.09$, $\sigma = 6.877$) from diverse backgrounds, one of whom dropped out of the study after the data collection ended, leaving us with a final total of 11 participants (see Table 1 in Appendix for details). Participants were recruited via posters advertising the study with on the university campus and via email on university mailing lists. The participants were split into two groups: 5 participants in

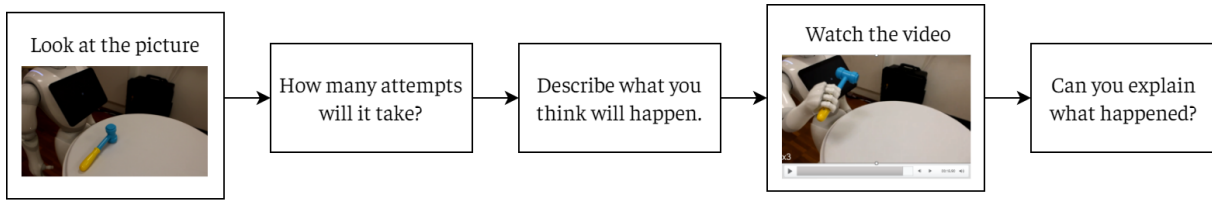
Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '24, May 11–16, 2024, Honolulu, HI, USA

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0331-7/24/05

<https://doi.org/10.1145/3613905.3651078>



group C1 (“mostly failure”) and 6 participants in group C2 (“mostly success”).

2.2 Overview

To capture what the dynamics behind the construction of a mental model of a robot we run a prediction game with two arms (“mostly success” vs. “mostly failure”) where the participant is asked to guess what the robot is going to do when tasked to pick up an item. The study was designed to induce the same initial beliefs about the robot’s capability in both arms prior to the start of the session and to vary the robot’s performance during the game.

We employed a think-aloud protocol to help discern whether the participants are using primed beliefs about the robot’s cognition or relying more directly on their experience during the session. In particular, we examine how experiencing a robot with a higher success rate vs. a robot with a lower success rate influences a) the prediction dynamics and b) how people make sense of the outcome. Intuitively, if a participant thinks that a robot “understands” the physical world it is attempting to manipulate, they should feel confident that the robot will succeed in novel tasks or that failures reflect the difficulty of the task. Presumably, *enough* failures might lead a participant to doubt the robot’s cognitive capacity. Similarly, enough positive experience might generate a belief that the robot has the corresponding cognitive capacity.

2.2.1 Training. After assigning the participant to experimental condition C1 or C2, we start the participant’s training phase: we show a sequence of slides that illustrate, via the use of videos and text, the robot learning how to pick up some objects via trial and error (see Section A.1.2 in Appendix for further details). This phase lasts for around 5 minutes and it is the same for all the participants, regardless of which condition they have been assigned to. The hope is that, by the end of the introduction, the participant is inclined to believe that the robot has recently learned to perform a simple task by itself and that it is capable of extending that skill to the handling of novel items. After the introduction, the participant is instructed on how the task will be performed during a familiarization round.

2.2.2 Data collection: Two types of data are collected during the session: the predicted number of attempts (quantitative data) and the participants’ reasoning (qualitative data). After the game, a semi-structured interview is also performed.

- **Predictions and Reasoning.** Figure 1 summarizes the experimental process. The participant is presented with the initial scene - a particular item is in front of the robot. Both before and after the outcome of the robot’s action, the participant is explicitly asked to (a) predict how many attempts will be

necessary for the robot to pick up an object or whether the robot will fail (a failure is considered when the robot takes 9 or more attempts before succeeding), and (b) voice their reasoning and justifications. In total, there are 8 rounds with objects of varying complexity and novelty. The interaction is audio-recorded and subsequently transcribed for qualitative analysis using the thematic analysis method.

- **Interview.** At the end of the session, a semi-structured interview is conducted in order to collect the participant general opinion on robots. The aim is to confirm that the perceived performance of the robot was concordant with the condition, to detect potential biases, to investigate the effect of anthropomorphism, and to speculate on the transferability of the robot’s skills. The interview is recorded, transcribed, and analysed in order to identify and extract the themes.

2.2.3 Analysis. We perform statistical analysis for small datasets on the quantitative data (predictions) in order to detect whether there is a statistically significant difference between the two conditions and to compare whether predictions between the conditions changed over time. The transcripts of the sessions and interviews are thematically analysed to extract possible patterns of reasoning of the participants while interacting with the robot (see Section A.1.1 in Appendix for further details).

3 FINDINGS

The resulting coding has been organized into two hierarchical structures. The first structure arranges the quotes on the base of item and condition. The idea is that it captures details of the evolving mental model as it undergoes dynamic changes and updates in both conditions (Table 2 in the Appendix). In contrast, the second structure is employed to ensure a clear understanding of the overall process of theorization (Table 4 in the Appendix). Both structures refer to the same quotes but arrange them differently: while in the first structure there is only one instance of each item (as it represents a comment of the participant in time), in the second structure some items were coded under multiple themes as they could belong to different patterns (e.g. Theory Topics and Reasoning Sophistication, etc.). The coding surfaced the following themes¹: (1) Theory Topics, (2) Reasoning Sophistication, (3) Anthropomorphisation of the robot, (4) Participant’s Beliefs, (5) Sensitivity to Errors.

3.1 Topics of Theorization / Theory Topics

When deliberating about the skills of the robotic agent, individuals made comments about the Cognitive capabilities 43% of the

¹Examples of quotes have been provided for each theme in Section B of the Appendix

time, Sensorial skills 36% of the time, and Mechanical skills 21% of the time. All the participants theorized about the robot's skills at least once, hence each item of the higher level of the hierarchy (Theory_cognitive, Theory_mechanical, Theory_sensorial) was coded in all the interviews (11), with the exception of the item Theory_mechanical, which was coded in 10 interviews. Overall, the coding between the two Conditions was evenly distributed. Figure 2 shows the number of references to Theory Topics over time, which appears to be decreasing.

3.1.1 Cognitive Skills. All the participants speculated about the robot's cognitive capabilities. The majority of comments (59%) were dedicated to analyzing the robot's "current" action. Among these, 62% portrayed the robot as employing a static approach with low cognitive skills, while 38% suggested that the robot could adapt to the situation, showcasing resilience and high cognitive skills. In contrast, 26% of the comments aimed at summarizing observations thus far and synthesizing a coherent "model" of the robot's cognitive skills. 15% of the time, (7) participants did not spontaneously provide any theorisation about the robot's actions. When questioned about the reasoning behind their prediction, 5 participants out of 7 replied they could not explain it (as presented in Section 3.2.1), while the others provided a theory.

3.1.2 Mechanical Skills. Ten (10) participants commented on the robot's mechanical capabilities with a particular focus on its hands. Overall, mentions of mechanical skills represented 21% of the total comments on Theory. The participants focused on the robot's force (9), robotic hand dexterity (8), body structure (3), speed (3), grip roughness (3) and Theory of Body (5). Theory of the Body (ToBy) included comments on the robot's physical awareness, and capacity to adapt to the surroundings.

3.1.3 Sensorial Skills. 36% of total Theory coding was about the robot's Sensorial Skills. Comments were divided into tactile (37%), vision (58%), and material (5%) which requires both senses in order to be identified. General observations on tactile and vision concerned the eventual presence of such sensory systems and where they would be located.

Comments on tactile concerned how the robot collects tactile information (proximity, touch, weight detection). Remarks on vision included observations on 2D (color, shapes) and 3D perspective. Seven (7) participants postulated that the robot is capable of understanding the item's stability, while 7 people commented on the robot's inability to detect a hollow space (items such as the cup, the shoe and the phone), which was almost univocally seen as a problematic item feature that the robot is blind to.

3.2 Reasoning Sophistication

The majority of people were able to fit the robot's successes and failures into a coherent narrative, though some participants felt "confused" at points during the interview and were unable to justify the robot's behavior. Overall, the coding between the two Conditions was evenly distributed. The different levels of sophistication are presented below from the least to the most complex.

3.2.1 Less Complex Reasoning. Some participants were not able to vocalize their reasoning when making predictions. This means that

they predicted an outcome without explaining why they thought so (see Section 3.1.1 for further details), and when asked what prompted them to make such a prediction, they admitted that they did not know. This happened sporadically to 5 participants. In two cases, this consistently occurred for a significant proportion of the study. 6 participants justified some of their predictions simply by stating that the robot would grasp the object in "the easiest way".

3.2.2 Simple Inductive Reasoning. Generally, the reasoning methods used to make a prediction focused on the object and the table setting, rather than the robot's capabilities. All 11 participants mentioned the features of objects when formulating predictions, rather than focusing explicitly on the robot's capabilities, indicating that they were making assumptions about the robot's abilities in an ego-centric way, that they did not fully verbalise. This was the code item with the highest number of comments in the Reasoning Sophistication theme. The features that were rated as most influential in how successful the robot's grasp was were the shape (how wide, how long, how "regular") and the position/orientation of the object. Some participants also mentioned the material, colour and the shading of the object. Sometimes participants (7) used previous outcomes or situations to justify their predictions rather than investigating the underlying reasons and deducing the robot's logical process.

3.2.3 Structured Reasoning. Ten (10) participants tried to understand the relationship between the robot's grasping technique and the object's features. Eight (8) participants refined this reasoning into a theory where the robot would identify two main models of objects (ball vs. bottle, or "short and large" vs. "tall and thin")² and develop a hierarchy experientially.

3.3 Anthropomorphism

Nine (9) participants compared the robot to a toddler in their discovery phase. Eight (8) participants commented at some point during the study that the robot does not know how to handle certain objects as a human would. When the participants (7)³ were asked if they thought that an industrial robot with the same intelligence would have performed better in the same setting, 3 participants responded that the outcome would have been the same, 1 participant replied that it depends on whether the mechanics of a human hand were transposed into the grasping limb, and 3 participants stated that the humanoid robot would certainly perform better as the objects used during these tasks were designed to be used by humans. Finally, when asked if the robot could transfer their skills to another task and context, all the participants (7)⁴ agreed that the robot could transfer its skills to another task, but that it would still require a period of initial training. This last item was coded in the Wrapping Up section of the First Coding Scheme (Table 2 in Appendix)

3.4 Participant's beliefs

Five (5) participants spontaneously commented that they believed that the robot was learning across the rounds. Four (4) participants

²The ball and the bottle were the two objects shown at the beginning of the study in the introductory slides as explained in Section A.1.2 of Appendix

³this question was asked only on 7 participants out of 11

⁴this question was asked only on 7 participants out of 11

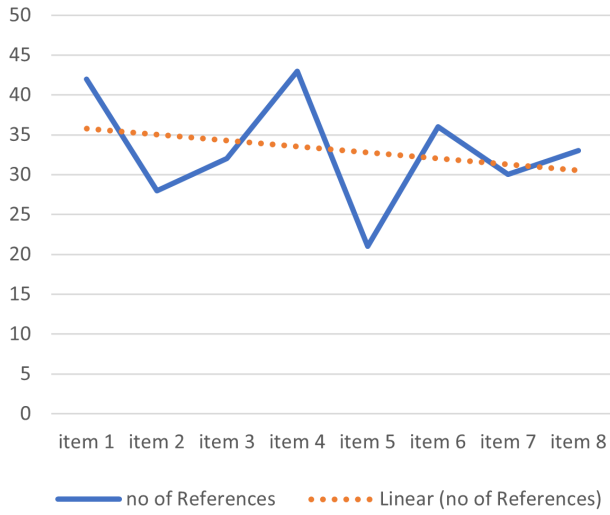


Figure 2: Number of References per item in Theory_Topics

theorized that the robot must have had some prior knowledge such as having witnessed a human performing the task, or that the robot must have had some pre-programming.

3.5 Error sensitivity

Participants tended to comment most (40%) after the outcome of the robot’s action confirmed their original prediction of Success, whereas they were commenting the least (12%) right after its opposite case (confirmation of the robot’s Failure).

The majority of the comments centered around either the Cognitive Limitations of the robot (primarily among participants in group C1) or the difficulty of the object in terms of shape, position, etc. (primarily among participants in group C2). Additionally, Mechanical Limitations and Sensorial Limitations were also referenced. In fewer instances, participants attributed the robot’s failure to being unlucky or suggested that it required more experience.

From the perspective of descriptive statistical analysis, it seems that, paradoxically, the participants who experienced the “most failures” condition (C1) were more likely to “trust” the robot and predict success. The group that experienced the robot mostly failing (C1), predicted Fail %22.22 of the total times, while the group that experienced the robot mostly succeeding (C2) predicted Fail %31.48 of the total times. We performed an N-1 Two Proportion Test (Two Tailed p-value: 0.3054519) which did not highlight a statistical difference between the percentages of the two groups. It may be that an effect was there, but not detected due to the small sample size, or it may be that there is no difference between the groups. If we only focus on the predictions of Success of the two groups and plot the values of the attempts that the participants predicted the robot would take, we can notice a growing trend in C1, as opposed to a steady trend in C2 (see Figure 3). This means that while the participants in C1 predicted more often that the robot would have been successful, they also expected each attempt to take longer compared to the C2 participants.

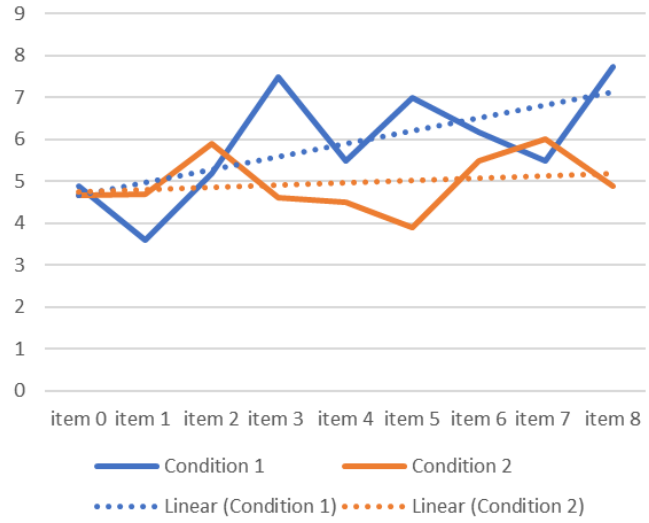


Figure 3: Trend of Success-predictions only.

4 DISCUSSION

General discussion on the results: In the upcoming section, we will delve into the outcomes arising from both thematic and descriptive analyses. Participants in condition C2, where the robot consistently succeeded, seemed to cultivate a heightened confidence in the robot’s learning abilities. These participants speculated as to whether the robot had been explicitly taught certain ways of grasping objects, which appeared notably human-like (e.g., holding a phone in the middle or gripping a hammer by the handle). This observation indicates that individuals in the C2 group struggled with conflicting thoughts, oscillating between viewing the robot purely as a machine and ascribing human-like thinking and behaviors to it. In contrast, C1 participants, where the robot often failed, commented less on these nuanced behaviors. The post-interaction interviews supported these findings.

Participants, when making comparisons, expressed expectations that the robot, with its humanoid physical structure, would perform comparably to, if not better than, an industrial robot. Moreover, when assessing the robot’s human-like intelligence, participants often likened it to a toddler. The reasoning behind this analogy was the robot’s resilient and curious approach to task discovery, underscoring participants’ inclination to humanize the robot. On a separate note, the findings suggest a prevalent focus on theorizing about the robot’s mental capabilities over its bodily attributes. This inclination may be linked to the robot’s anthropomorphic design, leading participants to intuitively engage with its human-like bodily features, often taking certain bodily capabilities (e.g. depth perception) for granted.

When theories were produced, participants in group C1 generated slightly more sophisticated theories than those in C2. A potential explanation is that varying levels of digital literacy contribute to differing levels of theory sophistication. Further analysis revealed that comments with high Reasoning Sophistication were

predominantly from P6, the sole participant with working experience in the technology field.

Over time, the data indicated a marginal decline in theorization frequency, hinting that individuals tend to swiftly settle on their initial theories—a phenomenon known as the anchoring effect in psychology. This is further confirmed by the fact that contrary to expectations, there was no surge in theorizing after a failure. On the contrary, people were more inclined to theorize immediately after their prediction of “Success” was confirmed by the outcome. This suggests that it might be easier to support a theory already in progress than to deconstruct a model constructed with difficulty.

A notable outcome of this study is that, contrary to our expectations, participants who witnessed the robot frequently failing (C1) were more inclined to “trust” the robot and predict success, despite the fact that the robot was not, in fact, successful. Possible explanations include a natural instinct to root for the underdog, wherein participants empathized with the robot and hoped for its success, or the gambler’s fallacy, where participants believed that after numerous failures, the probability of success in subsequent sessions was higher.

Comments on the methodology: In terms of methodology, a noteworthy observation is the wide range of theorization across participants, from minimal to extensive engagement. Exploring the correlation between the amount of discussion and the extent of theorization raises questions about causation: does increased conversation lead to more theorizing, or is the relationship between the two more nuanced? It is challenging to determine. All we can conclude is that the results show an absence of significant variance between conditions. All participants did spontaneously produce theories, although some individuals required prompting, some of the time.

Another key consideration is the apparent lack of a robust effect stemming from variations in error rates. Is there a specific setup circumstance that could strongly reinforce certain predictions and consequently lead to increased theorization? Determining if participants deeply reflect on the presented scenarios or if the impact is more surface-level is an interesting aspect to explore further.

Another question is whether this methodology is successful in HRI experiments where the cognitive status of the participants is a condition of the study, and we are interested in discovering not only the pre-existent participant’s beliefs but also if (and how) we can elicit a cognitive status. We used thematic analysis due to uncertainties we faced in previous studies [22, 23] where we were investigating the participant’s trust in robots based on the level of cognitive intelligence that the robot was believed to possess by the participants. The studies had two experimental conditions. One group of participants would watch a video showing the robot as not being fully cognitively developed, and vice versa, a second group would watch a video where the robot demonstrated being cognitively developed. During the interaction, all the participants experienced the same robot behavior (which lasted for a few rounds), so the only difference in their experience was limited to the initial conditioning. The results of these studies highlighted a statistical difference in the actions performed by the participants of the two groups during the interaction but did not show a significant difference in the questionnaire results about anthropomorphism,

likability, credibility, etc. This made us question whether the “conditioning” of the videos did not last for the whole duration of the experiment and whether the reiterated experience of the robot’s behaviour was more influential at the time the questionnaires were collected. But more importantly, it prompted us to explore whether it was possible to find a way to keep track of the belief status of the participants during the interaction.

A general conclusion of the current study is that using thematic analysis combined with a think-aloud protocol, when possible, does indeed allow access to participants’ beliefs and support tracking changes in their belief systems over time. It emphasizes the non-trivial nature of the cognitive conditioning process, indicating that imprinting a specific belief in people’s minds is highly dependent on both objective and subjective factors, as well as the timing and method of communicating the “conditioning” information.

Future Research Directions: One potential area for exploration is the impact of physical presence versus video presence during Human-Robot Interaction. The challenge arises when the experiment involves participants speculating about the robot’s performance, as there is a need to strike a balance between ensuring the environment remains controlled and ensuring the robot’s behavior is perceived as authentic. In using VR, we could simulate a real-life experience of physical presence while relinquishing control over the robot’s performance.

One limitation that arises from this design is the lack of statistical results, stemming from the small size of the dataset. To address this limitation, we plan to run a streamlined online version of the same game. Participants will interact with the robot’s videos using buttons and sliders. Our focus will be on collecting participants’ predictions, and a questionnaire will be administered at the end of the interaction. The results of this game will complement the findings of the current study, helping identify whether there are statistical differences in participants’ answers based on their diverse experiences with the robot’s performance.

5 CONCLUSION AND FUTURE WORK

In this research, we explored how individuals perceive robots, specifically examining how people develop theories about robots equipped with cognitive capabilities. Our focus is on understanding whether individuals attribute mental qualities to the robot and to what extent. To investigate this, we enlisted 12 participants in a guessing game where they observed a robot learning to manipulate objects. Subsequently, participants were tasked with predicting how the robot would perform with unfamiliar objects and speculating on its cognitive processes, including reasoning, sensing, and actuation. To assess the potential impact of the robot’s performance on the theorization process, participants were divided into two groups: one group observed a robot that mostly failed (C1), while the other group observed a robot that mostly succeeded (C2). We recorded participants’ speculations, transcribed them, and conducted a qualitative analysis to identify the predominant themes in their theories. The findings reveal intriguing patterns: participants, regardless of their condition, consistently engaged in theorizing about the robot in similar quantities and ways, which suggests that generating contrasting beliefs about the robot’s cognitive skills is not as straightforward as we thought. Surprisingly, those who

observed the robot failing more frequently were more inclined to offer the robot “another chance”. Additionally, the results provided valuable insights into potential limitations and enhancements for refining the protocol aimed at evaluating human theories about their robotic companion.

REFERENCES

- [1] Anne Adams, Peter Lunt, and Paul Cairns. 2008. A qualitative approach to HCI research. In *Research Methods for Human-Computer Interaction*. Paul Cairns and Anna Cox (Eds.). Cambridge University Press, Cambridge, UK, 138–157. <http://www.cambridge.org/catalogue/catalogue.asp?isbn=9780521870122&ss=toc>
- [2] Cindy L. Bethel, Jessie E. Cossitt, Zachary Henkel, and Kenna Baugus. 2020. *Qualitative Interview Techniques for Human-Robot Interactions*. Springer International Publishing, Cham, 145–174. https://doi.org/10.1007/978-3-030-42307-0_6
- [3] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.
- [4] Evgeniy Bryndin. 2020. Formation of technological cognitive reason with artificial intelligence in virtual space. *Britain International of Exact Sciences (BloEx) Journal* 2, 2 (2020), 450–461.
- [5] Alejandro Chacón, Pere Ponsa, and Cecilio Angulo. 2021. Cognitive interaction analysis in human–robot collaboration using an assembly task. *Electronics* 10, 11 (2021), 1317.
- [6] Min Chen, Stefanos Nikolaidis, Harold Soh, David Hsu, and Siddhartha Srinivasa. 2018. Planning with Trust for Human-Robot Collaboration. In *Proceedings of the 2018 ACM/IEEE International Conference on Human-Robot Interaction* (Chicago, IL, USA) (HRI '18). Association for Computing Machinery, New York, NY, USA, 307–315. <https://doi.org/10.1145/3171221.3171264>
- [7] Vivienne Bihe Chi and Bertram F. Malle. 2023. People Dynamically Update Trust When Interactively Teaching Robots. In *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction* (<conf-loc>, <city>Stockholm</city>, <country>Sweden</country>, </conf-loc>) (HRI '23). Association for Computing Machinery, New York, NY, USA, 554–564. <https://doi.org/10.1145/3568162.3576962>
- [8] Alejandra Ciria, Guido Schillaci, Giovanni Pezzulo, Verena V Hafner, and Bruno Lara. 2021. Predictive processing in cognitive robotics: a review. *Neural Computation* 33, 5 (2021), 1402–1432.
- [9] Chien Dang, Heungju Ahn, Hyeon Seo, and Sang C Lee. 2020. A Cognitive Robotic System for a Human-Following Robot. *Frontiers in Artificial Intelligence and Applications* 331 (11 2020), 601–607. <https://doi.org/10.3233/FAIA200737>
- [10] Giacomo De Rossi, Marco Minelli, Alessio Sozzi, Nicola Piccinelli, Federica Feraguti, Francesco Setti, Marcello Bonfé, Cristian Secchi, and Riccardo Muradore. 2019. Cognitive robotic architecture for semi-autonomous execution of manipulation tasks in a surgical environment. In *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, Macau, China, 7827–7833.
- [11] Maged Farouk. 2022. Studying Human Robot Interaction and Its Characteristics. *International Journal of Computations, Information and Manufacturing (IJCIM)* 2, 1 (2022), 38–49.
- [12] Kevin Anthony Hoff and Masooda Bashir. 2015. Trust in automation: Integrating empirical evidence on factors that influence trust. *Human factors* 57, 3 (2015), 407–434.
- [13] Zahra Rezaei Khavas. 2021. A Review on Trust in Human-Robot Interaction. [arXiv:2105.10045 \[cs.RO\]](https://arxiv.org/abs/2105.10045)
- [14] Bing Cai Kok and Harold Soh. 2020. Trust in robots: Challenges and opportunities. *Current Robotics Reports* 1 (2020), 297–309. <https://doi.org/10.1007/s43154-020-00029-y>
- [15] Bimal Kumar and Akash Dutt Dubey. 2017. Evaluation of trust in robots: A cognitive approach. In *2017 International Conference on Computer Communication and Informatics (ICCCI)*. IEEE, Coimbatore, India, 1–6.
- [16] Danny Lange. 2019. Cognitive robotics: Making robots sense, understand, and interact. *Computer* 52, 12 (2019), 39–44.
- [17] Sukhan Lee and Ahmed M Naguib. 2020. Toward a sociable and dependable elderly care robot: design, implementation and user study. *Journal of Intelligent & Robotic Systems* 98 (2020), 5–17.
- [18] Séverin Lemaignan, Mathieu Warnier, E Akin Sisbot, Aurélie Clodic, and Rachid Alami. 2017. Artificial cognition for social human-robot interaction: An implementation. *Artificial Intelligence* 247 (2017), 45–69.
- [19] Michael Lewis, Katia Sycara, and Phillip Walker. 2018. *The Role of Trust in Human-Robot Interaction*. Springer International Publishing, Cham, 135–159. https://doi.org/10.1007/978-3-319-64816-3_8
- [20] Bertram F. Malle, Eric Rosen, Vivienne B. Chi, and Dev Ramesh. 2023. What Properties of Norms can we Implement in Robots?. In *IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. IEEE, Busan, Korea, 2598–2603. <https://doi.org/10.1109/RO-MAN57019.2023.10309355>
- [21] Adnan Masood, Adnan Hashmi, Adnan Masood, and Adnan Hashmi. 2019. Cognitive robotics process automation: Automate this! *Cognitive Computing Recipes: Artificial Intelligence Solutions Using Microsoft Cognitive Services and TensorFlow* 1 (2019), 225–287.
- [22] Wenxuan Mou, Martina Ruocco, Debora Zanatto, and Angelo Cangelosi. 2020. When Would You Trust a Robot? A Study on Trust and Theory of Mind in Human-Robot Interactions. In *IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. IEEE, Naples, Italy, 956–962.
- [23] Martina Ruocco, Wenxuan Mou, Angelo Cangelosi, Caroline Jay, and Debora Zanatto. 2021. Theory of mind improves human’s trust in an iterative human-robot game. In *Proceedings of the 9th International Conference on Human-Agent Interaction*. ACM, Virtual event, Japan, 227–234.
- [24] Shingo Shimoda, Lorenzo Jamone, Dimitri Ognibene, Takayuki Nagai, Alessandra Sciutti, Alvaro Costa-Garcia, Yohei Oseki, and Tadahihiro Taniguchi. 2022. What is the role of the next generation of cognitive robotics? *Advanced Robotics* 36, 1-2 (2022), 3–16.
- [25] Tadahihiro Taniguchi, Daichi Mochihashi, Takayuki Nagai, Satoru Uchida, Naoya Inoue, Ichiro Kobayashi, Tomoaki Nakamura, Yoshinobu Hagiwara, Naoto Iwashashi, and Tetsunari Inamura. 2019. Survey on frontiers of language and robotics. *Advanced Robotics* 33, 15-16 (2019), 700–730.
- [26] Takane Ueno, Yuto Sawa, Yeongdae Kim, Jacqueline Urakami, Hiroki Oura, and Katie Seaborn. 2022. Trust in Human-AI Interaction: Scoping Out Models, Measures, and Methods. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. ACM, New Orleans, L.A., 1–7. <https://doi.org/10.1145/3491101.3519772>
- [27] Jian Zhao, Mengqing Wu, Liyun Zhou, Xuezhong Wang, and Jian Jia. 2022. Cognitive psychology-based artificial intelligence review. *Frontiers in Neuroscience* 16 (2022), 1024316.

A APPENDIX

A.1 Materials and Methods: supplementary material

Table 1: Demographics of the participants. Participants with even IDs are from group C1 (most failures), and vice versa for the participants with odd numbers.

Condition	ID	Background	Gender	Age
C1	P0	business	M	41
C2	P1	marine science	M	22
C1	P2	medicine	M	20
C2	P3	business	M	23
C1	P4	illustrator	F	32
C2	P5	marine science	F	24
C1	P6	linguistics, IT	M	38
C2	P7	history	M	25
C1	P8	marine science	F	30
C2	P9	teacher	F	23
C2	P11	filmmaking; nutrition	F	31

A.1.1 Thematic Analysis. Given the potential complexities of belief formation and maintenance for such sophisticated assessments, rather than using a predefined set of codes deduced from our hypothesis, we decided to have an open and inductive reasoning approach. This means that we started by reading the transcriptions without preconceptions, we tried summarizing and categorizing the participants’ statements and then cross-referenced the codes between the different interviews multiple times until a coherent framework emerged.

The coding has been handled by the first author over the course of three passes. The first coding was done with pen and paper in order to break down the transcript and surface an initial coding structure. In the second phase, the transcripts were analysed in

NVIVO, where the coding structure went through a significant restructuring and reached a hierarchical multi-level form that organizes the quotes based on the stage of the interview (before item 1, after item 1, etc) and the group condition. We called this the First Coding Scheme (Table 2).

In the third and last phase, the coding was duplicated and re-structured once again re-grouping the codes that surfaced at the earlier stage in macro-codes. We called this Second Coding Scheme (Table 4). Both the coding structures have been used to produce the findings, as the first one retained information about the dynamic changes and updates the mental model was going through over time in the two conditions, while the second is used to have a clear vision of the general theorization process.

A.1.2 Training Slides. The slides start by stating that the robot had been trained to autonomously learn how to pick up objects:

Slide 1: “We are currently working on a way a robot can learn in an experiential way. Basically, what happens is that the robot tries to learn to achieve a goal by performing (and especially failing) several actions.”

Following, we show the participant a series of videos of the robot in its training phase:

Slide 2: “In our case, our robot learned how to pick up different objects, for example, a bottle and a ball.”
* video of robot picking up a bottle and a ball * (See Figure 4)

Slide 3: “This is how the robot learns how to pick up a ball!”

* video of the robot’s many attempts at learning how to pick up a ball *

Slide 4: “This is how the robot learns how to pick up a bottle based on what it learned from picking up a ball!”

* video of the robot’s many attempts at learning how to pick up a bottle *

Finally, the rules of the game are introduced:

Slide 5: “Let’s play a game. I will show you a sequence of objects and I’d like you to tell me if you think Pepper will be able to grasp the object in 9 attempts or less.”

Familiarization round: At this point, the participant is shown an object (a hammer toy) and is asked (a) to predict how many attempts will be necessary for the robot to successfully grasp such object, and (b) to voice the reasoning behind their prediction. See Figure 1 for reference.

Slide 6: “Now, based on what you saw, what do you think will happen if Pepper tries to pick up a hammer? And why? And how many attempts do you think it takes?”

* image of a hammer toy on a table and the robot behind it *

Here, the participant casts their prediction and explains their reasoning behind such prediction. After that, they watch the outcome:

Slide 7: * video of robot learning how to pick up a hammer *

Finally, the participant is asked to provide a comment on the outcome:

Slide 8: “Success! Pepper managed to grab the hammer in 3 attempts! Why do you think that happened?”

A.2 Coding Schemes

The First coding scheme is presented in Table 2. Each “Condition” is then further split into items (see Table 3).

The Second coding scheme is presented in Table 4.

We don’t report the reference numbers for each item as it won’t provide any additional information.

B PARTICIPANTS’ QUOTES

B.1 Theory_total

B.1.1 Cognitive Skills.

#explanation of phenomena: #static_approach

P0: *Um, he wasn’t, the adjustment wasn’t that much. It was more like he was coming directly to the object and not thinking of how the object was displayed. I didn’t see... different angles or attempts of rotating the arm, or the hand or the fist, trying to go in a different way. So he was dealing with the shoe as a whole object, regardless of how the shoe was displayed or placed on the table or where it was facing.*

#explanation of phenomena: #dynamic_approach

P5: *I don’t think it was a matter of knowledge as much as it just kind of adjusted the angle at which it’s picking. So at first, the fingers kind of went in the centre of the cube, then a little bit further out, and then finally it sort of did adapt properly. So not knowledge as much, just experience. I think it just learns from the first attempt and then...*

#abstraction of theory:

P3: *Everything is— it might be from the dimension of the object or the side of the object. So it might like have with the sensors, you know, checks the size of an object and then it might take them most optimal point that it finds to carry the object.*

#no theory_no explanation

P5: *I want to say maybe four attempts, and it’ll try to pick it up from the handle.*

B.1.2 Mechanical Skills.

#ToBy (Theory of the Body):

P9: *Yeah. It knows where it’s own hand is. He knows if I go in sideways, that’s that will be better to grip things. So it must have an awareness of where it’s own arm is to be able to put in the right place.*

#force:

P4: *I think he just needs to work on his grip strength. He—yeah, it was like he was making good attempts but it was like he wasn’t gripping hard enough.*

#dexterity:

P9: *He does have the dexterity in his fingers that I didn’t think he had. [...] He’s learnt to use his fingers more than his palms. Before it was a very whole hand event. Now it’s just fingers a bit more. Because it’s smaller and I guess he knows that he has to just be a bit more gentle with what he’s doing with his hands.*

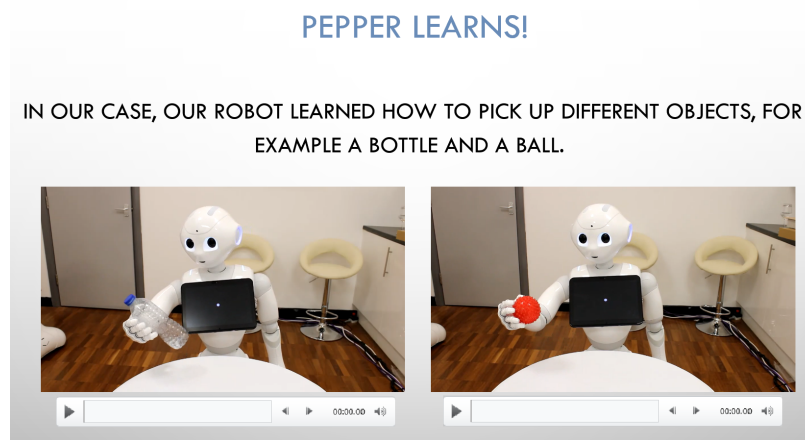


Figure 4: Slide 2 of the robot training presentation. The robot picks up a bottle (left) and a ball (right).

Table 2: First coding scheme. Coding carries information about the timing of comments. Percentages are expressed in a relative manner: (*) Before Predicting sub-categories % are in relation to the total amount of Before Predicting comments (n=291); (**) After Predicting sub-categories % are in relation to the total amount of After Predicting comments (n=275); Condition % are in relation to the total amount of comments in their parent

Code	Files	References	Percentages
0_During Training	3	10	
1_Before Predicting Fail	11	94	32% (*)
Condition 1	5	51	54%
Condition 2	6	43	46%
1_Before Predicting Success	11	197	68% (*)
Condition 1	5	106	54%
Condition 2	6	91	46%
2_After Correct Prediction Fail	9	33	12% (**)
Condition 1	4	22	67%
Condition 2	5	11	33%
2_After Correct Prediction Success	11	109	40% (**)
Condition 1	5	40	37%
Condition 2	6	69	63%
2_After Predicting Fail but it's Success	9	52	19% (**)
Condition 1	3	9	17%
Condition 2	6	43	83%
2_After Predicting Success but it's Fail	11	81	29% (**)
Condition 1	5	53	65%
Condition 2	6	28	35%
3_Wrapping up	11	99	
Condition 1	5	46	46%
Condition 2	6	53	54%

B.1.3 Sensorial Skills.

P1: So obviously I'd say it's definitely got some sort of camera sensor so it can see the object and... sort of recognize its—the directions of the object. I would guess it's probably also got some sort of pressure sensors in the hand so that it can tell when it's gripping onto an object, because heavier objects it seemed to be a lot better at picking up than—for example, the black cup, it was like it would just slide out of the way instantly.”

#interaction with objs:

P1: I would guess it's probably also got some sort of pressure sensors in the hand so that it can tell when it's gripping onto an object, because heavier objects it seemed to be a lot better at picking up than—for example, the black cup, it was like it would just slide out of the way instantly.”

#stability:

Table 3: Structure underlying Condition 1 and Condition 2 of the First coding scheme.

item	Condition 1	Condition 2
0	hammer	hammer
1	cube	lint roll
2	lint roll	green cup
3	green cup	adapter
4	phone	lightbulb
5	adapter	phone
6	shoe	black cup
7	lightbulb	shoe
8	black cup	cube

P1: I do think it could understand whether the objects were stable, because like the example of the light bulb, to begin with it was knocking it over a lot and it seemed to be making—like adjusting to be a bit more delicate as it was picking it up, the more it attempted to pick it up, until it obviously succeeded at the end where it didn't knock it over."

#hollow:

P4: So I don't know if he sees like the depth of the—like inside of the cup. I don't know. Like, I don't know how good his perception is of like 3D objects. So like, it would—so if I was trying to pick up the cups then I would be able to see immediately that there's a big dip in the middle that I could just grasp onto. But I don't know if he maybe sees it as like just a 3D object with like a flat top and just like sides, like just a cylinder shape. So I think that would be interesting to see if he's able to like grab it from the side like a normal person would, or if he's able to understand that it's got an inside bit that can be grasped."

B.2 Reasoning Sophistication

B.2.1 Less Complex Reasoning.

#Participant doesn't know:

INT: Where—do you think—what do you think—what kind of sensorial information do you think the robot perceives?

P8: Oh, okay... I reckon it perceives.... Oh, I think—this is really hard"

#It's just easier:

P7: I think I'm going to stay with my previous thought about that he rather go for the top things than movement of the elbow. It's not quite perfect yet, so it is just easier I think, for him just to grab it by the top bits. [...] It requires less movements."

B.2.2 Simple Inductive Reasoning.

#Object features:

P0: Depending on the shape, depending on the weight and depending on... where the object is placed. All the objects that we were dealing with right now were at chest level on the table, so if the object was lower or higher, it will be a different case for the robot because the angle will definitely change. Some objects will be easier to lift from a lower level, others will be more difficult.

#Comparison with previous objects:

P1: I mean, having seen it struggle with the ball and the bottle to begin with, which is like to me I'd assume a relatively simple shape." From another interview:

INT: why did you think it would have attempted from the sides?

P5: Because he's done that with the other objects when the approach didn't work."

B.2.3 Structured Reasoning.

#Robot analyses the objects:

P1: Cool. So with this cup it realized that it had to come side-on rather than top-down, unlike the first cup. Which I would guess is, from the way I've seen it working, probably to do with the fact that it's a taller and narrower cup so the robot knew to go for the longer narrower edge, which it seems to have done with some of the other objects."

#Two Models:

P6: I assume that it's kind of iterative, and that creates variations. Like in—almost genealogy. In like, we got the original two—Adam and Eve, the ball and the bottle. And that becomes like, tube and the solid object. And then from that, it kind of diverts into different positions, interactions with the table position, or something like that. And then even there, like, there might be cross-breeding between the models, to kind of—when there's like, success. Like, it picks up an object that has both characteristics from both models in terms of shape, or position. And those models interact. And there's—again, another branch of the tree is born based on those interactions. And the characteristics from both models that got adopted into the new model. Yeah, that's how I can visualize it being kind of..."

B.3 Anthropomorphism

#Like a child:

INT: So you said [...] during the viewing of the video—of the clip, "Oh, it's just like a baby!" [...] Why did you say that?

P4: Because they'll kind of like slip and then just immediately try again. [...] it's almost as [...] they're obviously processing the information but there's no like emotional reaction to it, so whereas like if an adult was trying to do something lots and lots of times they'd get annoyed with themselves. Or from what I've seen, a lot of adults get annoyed with themselves if they keep like repeatedly get things wrong. Whereas with babies, and like toddlers, they just keep going until they get it and it kind of reminded me of that."

#Grasping (not) like a human:

P9: I think it knew that the middle wasn't even, that it did have different circumferences. And it went for the bottom [roll] rather than the top [handle]. Because I guess it doesn't know which bits the handle because it doesn't have the context of objects. So the bottom is perfectly easy to pick up, but if you actually wanted to use the lint roll you'd have to pick it up by the top [handle]. But I guess it doesn't know that."

#Industrial vs humanoid:

P6: I think the humanoid one at least has the advantage in interacting with kind of human-centric objects. Because it kind of had the fingers, and it can go around objects, like when grasped the adapter, was kind of just with a couple of fingers. So I would say that given the same programming, I think this one would still kind of win, because it has kind of—a bit more dexterity than, like, a industrial robot with just a clamp, or something like that. Maybe I'm wrong, maybe the simplicity would play in its favour."

#Transferable skills:

Table 4: Second coding scheme. “N” refers to the number of files where each code is present (11 max). “Ref” are the number of quotes coded under each code. C1 and C2 are respectfully “mostly fail” and “mostly success” conditions. “Refs.(%)” are the percentages for each item with respect to the item’s parent. In “%(C1)” and “%(C2)” we report the percentages for each item for each condition.

Code	N	Ref	%	N(C1)	Ref(C1)	%(C1)	N(C2)	Ref(C2)	%(C2)
Total	11	662	100%	5	324	49%	6	338	51%
Anthropomorphism	10	40	6%	5	23	58%	5	17	43%
Grasping objs (not) like a human	8	14	35%	4	7	50%	4	7	50%
Industrial vs humanoid	7	10	25%	3	5	50%	4	5	50%
Like a child	9	16	40%	5	11	69%	4	5	31%
Participant’s belief	9	28	4%	3	6	21%	6	22	79%
Evidence of rbt learning	5	11	39%	2	4	36%	3	7	64%
Preprogrammed or taught	5	19	68%	1	4	21%	4	15	79%
Error Sensitivity	11	84	13%	5	55	65%	6	29	35%
Object too difficult	9	23	27%	3	9	39%	6	14	61%
Mechanical limitations	7	13	15%	4	8	62%	3	5	38%
Sensorial Limitations	7	12	14%	4	9	75%	3	3	25%
Cognitive Limitations	7	25	30%	5	20	80%	2	5	20%
Unlucky	3	3	4%	2	2	67%	1	1	33%
Not enough experience	3	7	8%	2	6	86%	1	1	14%
Reasoning Sophistication	11	167	25%	5	80	48%	6	87	52%
Less Complex Reasoning	8	31	19%	5	16	52%	3	12	39%
It’s just “easier”	6	13	42%	3	4	31%	3	9	69%
Participant doesn’t know	5	18	58%	3	12	67%	2	6	33%
Simple Inductive Reasoning	11	92	55%	5	41	45%	6	51	55%
Comparison with previous objects	7	22	24%	3	11	50%	4	11	50%
Object features	11	70	76%	5	30	43%	6	40	57%
Structured Reasoning	10	44	26%	4	23	52%	6	21	48%
Robot analyses the objects	10	44	100%	4	23	52%	6	21	48%
# Two Models	8	27	61%	3	18	67%	5	9	33%
Theory_total	11	343	52%	5	160	47%	6	183	53%
Theory_cognitive	11	146	43%	5	63	43%	6	83	57%
abstraction of theory	10	38	26%	4	23	61%	6	15	39%
explanation of phenomena	11	86	59%	5	36	42%	6	50	58%
# dynamic_approach	9	33	38%	4	16	48%	5	17	52%
# static_approach	10	53	62%	5	20	38%	5	33	62%
no theory_no explanation	7	22	15%	2	4	18%	5	18	82%
Theory_mechanical	10	73	21%	5	36	49%	5	37	51%
calibration	1	2	3%	1	2	100%	0	0	0%
dexterity	8	24	33%	4	14	58%	4	10	42%
force	9	26	36%	5	14	54%	4	12	46%
grip	3	3	4%	2	2	67%	1	1	33%
speed	3	4	5%	0	0	0%	3	4	100%
structure	3	8	11%	1	3	38%	2	5	63%
ToBy	5	6	8%	1	1	17%	4	5	83%
Theory_sensorial	11	124	36%	5	62	50%	6	62	50%
material	6	7	6%	2	2	29%	4	5	71%
tactile	11	43	35%	5	22	51%	6	21	49%
# interaction with objs	7	16	37%	3	7	44%	4	9	56%
# weight	5	9	21%	1	3	33%	4	6	67%
vision	11	74	60%	5	38	51%	6	36	49%
# color	5	18	24%	3	11	61%	2	7	39%
# vision_3D	9	34	46%	3	13	38%	6	21	62%
## hollow	7	14	41%	3	8	57%	4	6	43%
## stability	5	7	21%	1	1	14%	4	6	86%

“INT: so can you imagine this robot having transferable skills? Like do you think this robot can transfer the things that it learned to another task easily?”

P7: I think the baseline can be transferred but he will need to -as with the shoe- he will need to know what he’s dealing with, trying to explore first and then... start rewriting the baseline of the previous hand-based movement.”

This last item was coded in the Wrapping Up section of the First Coding Scheme.

B.4 Participant’s beliefs

#Evidence of rbt learning

“P0: he... was doing better every single time so it was clear that there was progress in how he was trying to pick it up and he was learning”

#Preprogrammed or taught

“INT: why do you think it’s going to try in the middle? [...]”

P3: I think I might be because it’s a performance action that it was taught to or seen [...] Because not everyone would carry the phone

from the middle like, like we used to carry it from the bottom like since, I don’t know, like childhood.”

From another interview:

“INT: What do you think is the autonomy level of this robot from one to ten?”

[...]

P11: One. [...] because I think he did adjust some things depending on the object like I said before [...]. But then, I mean... it is programmed and the instructions are given. Well, let’s say two maybe. Maybe one is low.”

B.5 Error sensitivity

#Cognitive Limitations

P0: the adjustment wasn’t that much. It was more like he was coming directly to the object and not thinking of how the object was displayed. I didn’t see...different angles or attempts of rotating the arm, or the hand or the fist, trying to go in a different way. So he was dealing with the shoe as a whole object, regardless of how the shoe was displayed or placed on the table or where it was facing.